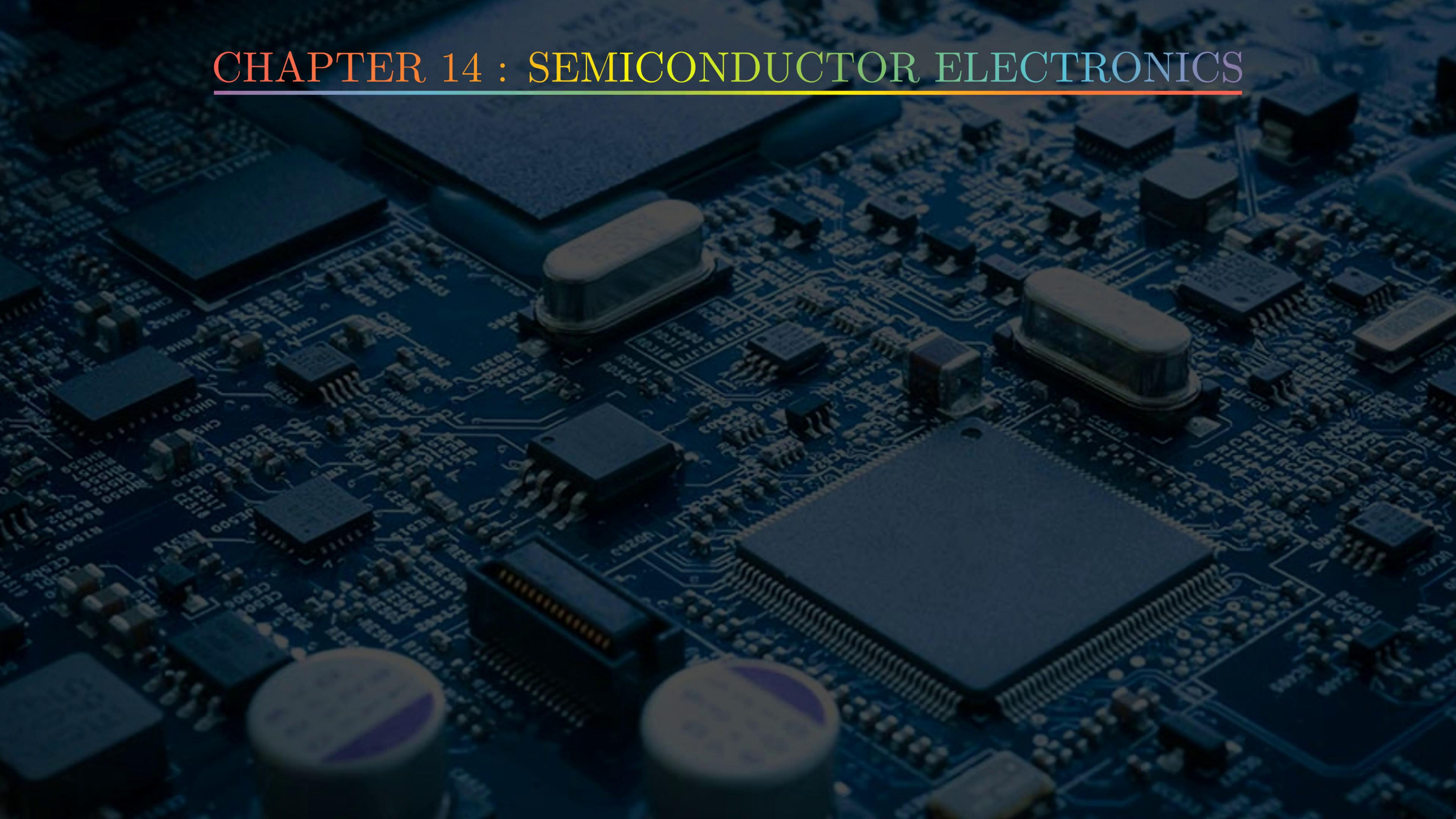


CHAPTER 14 : SEMICONDUCTOR ELECTRONICS

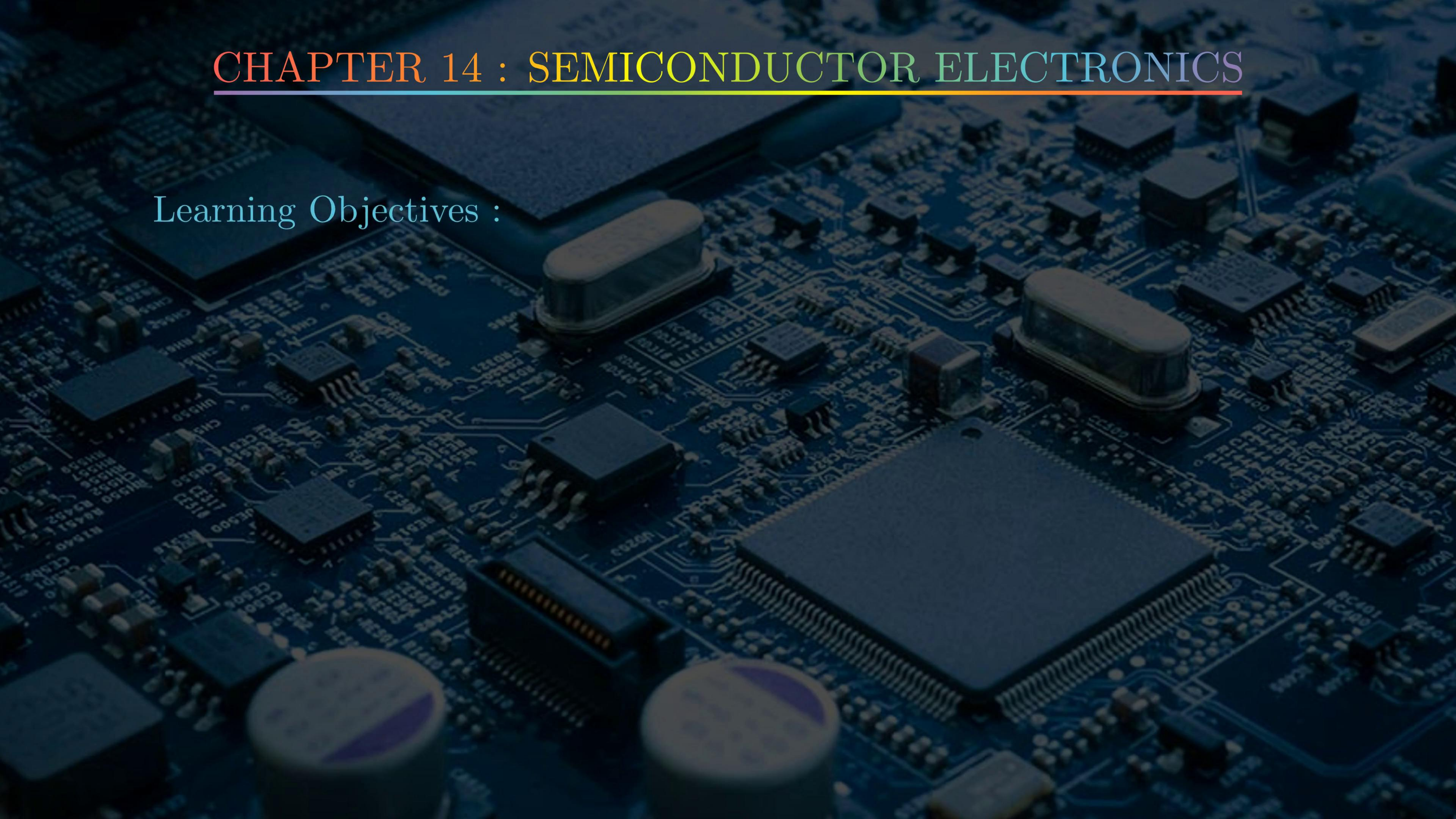


CHAPTER 14 : SEMICONDUCTOR ELECTRONICS

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- I-V characteristics in forward and reverse bias
- Application of junction diode - diode as a rectifier.

Classification of Metals, Insulator and Semiconductor

Semiconductor Electronics: Material Devices and Simple circuit

ON THE BASIS OF CONDUCTIVITY

(1) For metals:

$$S \sim 10^2 - 10^8 \Omega m$$

$$\sigma \sim 10^2 - 10^8 S/m$$

They have high conductivity.

(2) For Semiconductors:

$$S \sim 10^{-5} - 10^6 \Omega m$$

$$\sigma \sim 10^5 - 10^6 S/m$$

They have intermediate conductivity to metals and insulators.

(3) For insulators:

$$S \sim 10^{11} - 10^{19} \Omega m$$

$$\sigma \sim 10^{-11} - 10^{-19} S/m$$

They have low conductivity

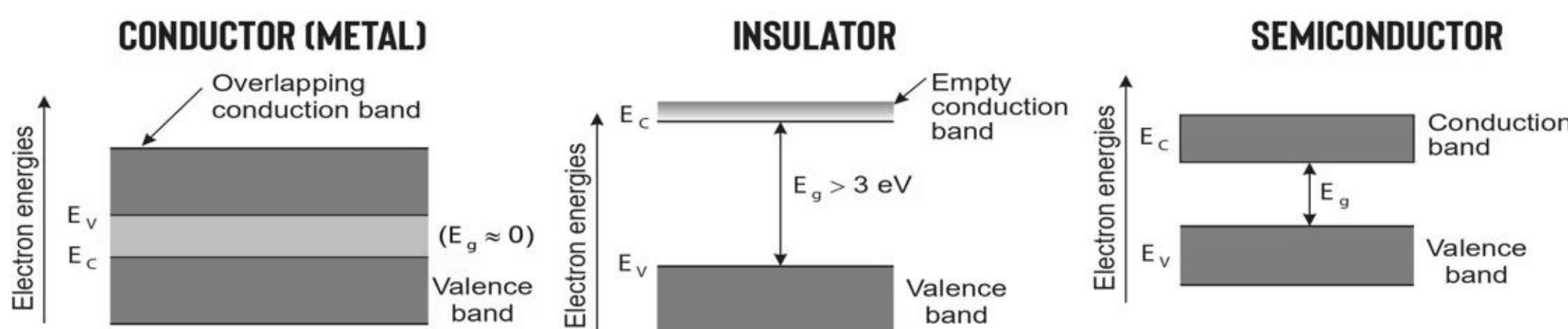
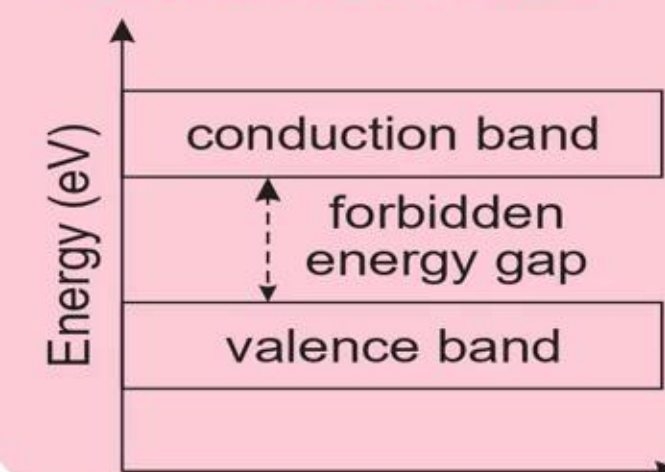
σ = electrical conductivity

ρ = resistivity

The band which is completely filled with electrons at OK is called valence band.

Conduction Band is completely empty at OK.

Energy band gap is the difference between valence band and conduction band



SEMICONDUCTOR AND ITS TYPES

Semiconductors exhibit electrical conductivity between conductors and non-conductors.

INTRINSIC SEMICONDUCTORS

- (1) Pure Semiconductors are intrinsic Semiconductors.
- (2) $n_i = n_e = n_h$, where, n_e = No. of free electrons, n_h = No. of holes, n_i = intrinsic carrier Concentration
- (3) Examples:- Ge, Si

EXTRINSIC SEMICONDUCTORS

- (1) Impure or doped Semiconductors are said to be extrinsic Semiconductors
- (2) Impurities are added to improve Conductivity

N - type Semiconductor $n_e \gg n_h$

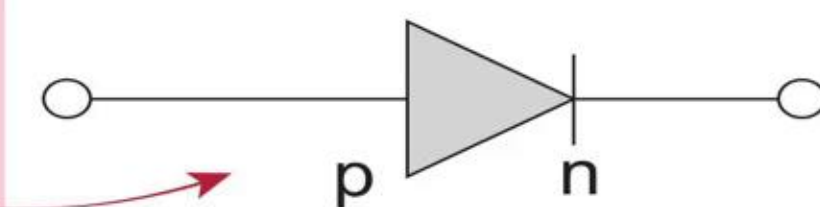
- (1) Electrons are majority charge carriers.
- (2) Holes are minority charge carriers.
- (3) Si or Ge doped with Pentavalent elements (P, As, Sb)

P - type Semiconductor $n_h \gg n_e$

- (1) Si or Ge doped with trivalent (B, Al) elements
- (2) Electrons are minority charge carriers.
- (3) Holes are majority charge carriers

THERMAL EQUILIBRIUM

The electron and hole Concentration in a Semiconductor in thermal equilibrium is given by, $n_e n_h = n_i^2$



P - N JUNCTION diode

P - N junction diode is the combination of P - type and N - type Semiconductor.

P - region has mobile majority holes and immobile - ve ions.

N - region has mobile majority free electrons and immobile positively charged ions.

POTENTIAL BARRIERS

Potential barrier is the potential difference developed across depletion region.

$$V_B = 0.7 \text{ for Silicon}$$

$$= 0.3 \text{ for germanium}$$

FORWARD BIAS

In Forward Bias

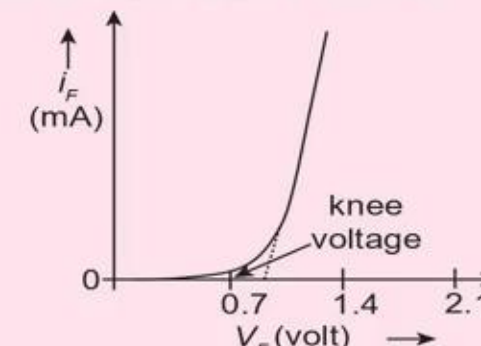
- (1) +ve terminal to P - Side
- (2) -ve terminal to N - Side
- (3) depletion layer reduced
- (4) diffusion current increases

REVERSE BIAS

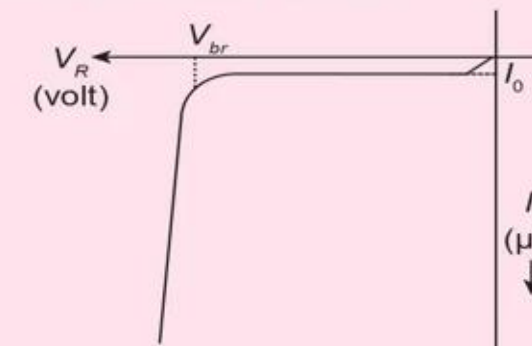
In Reverse Bias

- (1) -ve terminal to P - Side
- (2) +ve terminal to N - Side
- (3) depletion layer increases
- (4) diffusion current increases

Forward characteristics curve



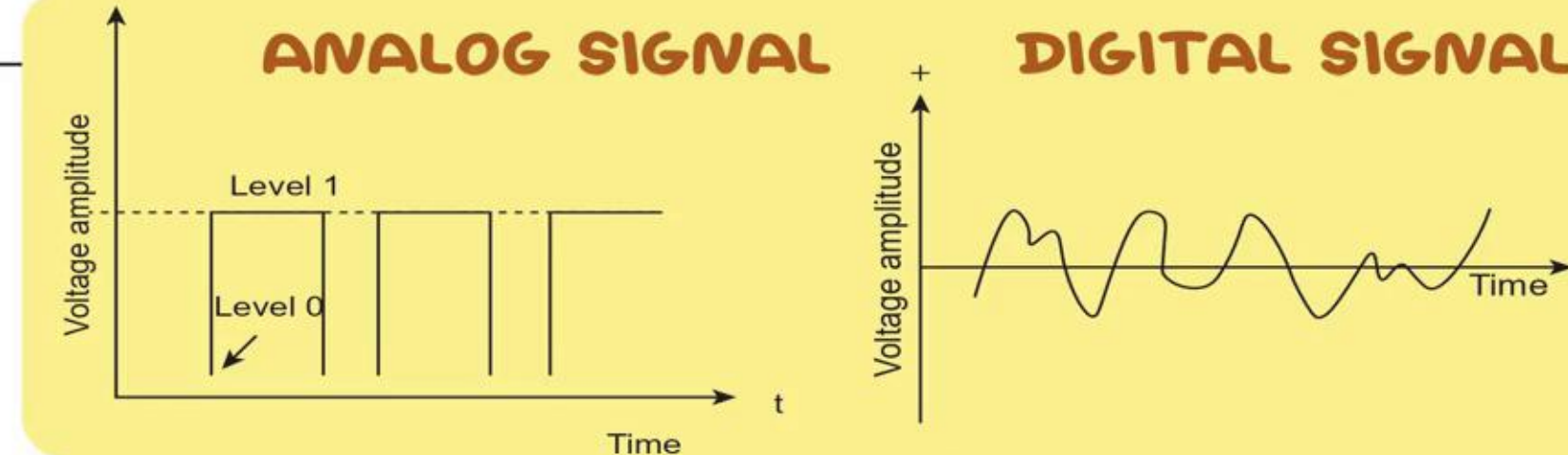
Reverse characteristic curve



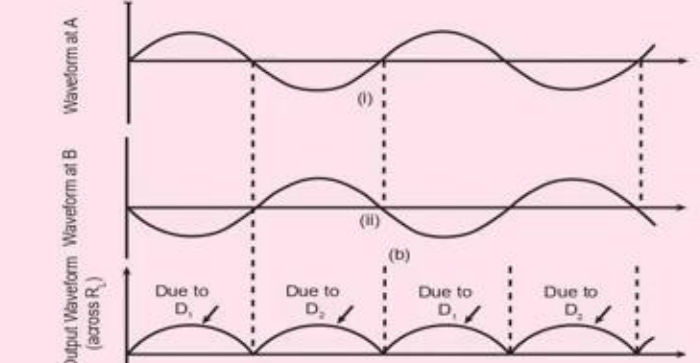
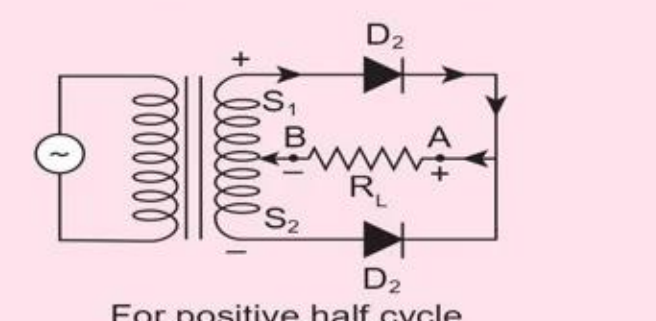
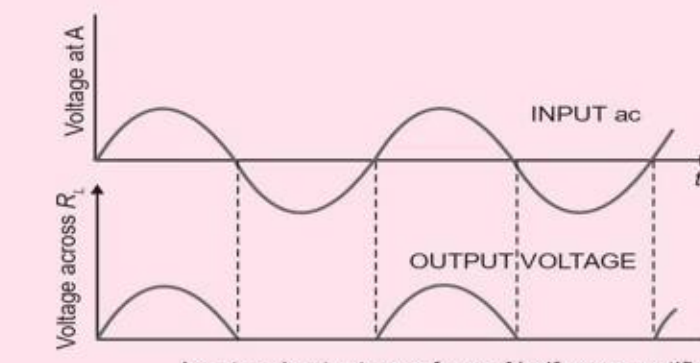
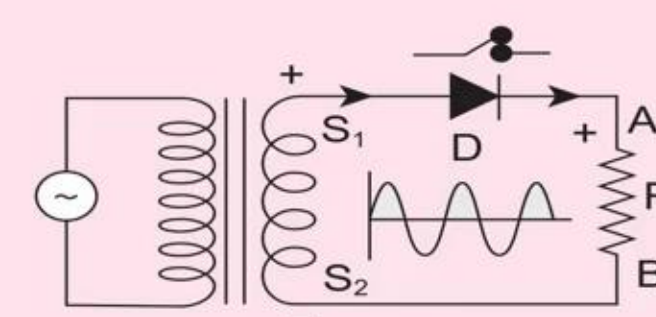
TRANSISTOR

Transistor is a three terminal device

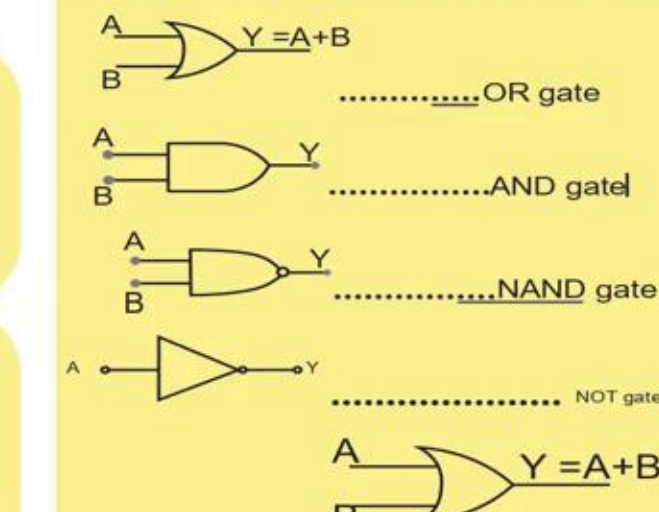
- (1) Emitter (E)
- (2) Base (B)
- (3) Collector (C)



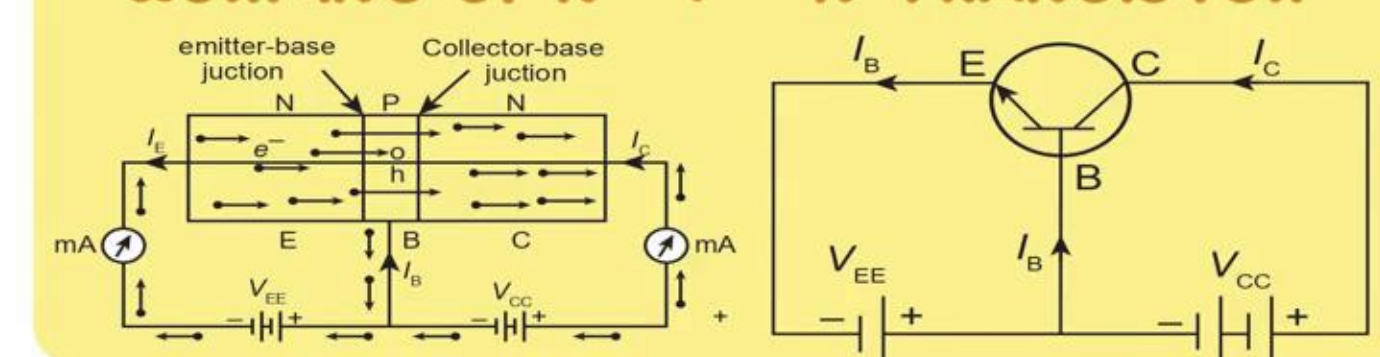
APPLICATIONS OF JUNCTION DIODE



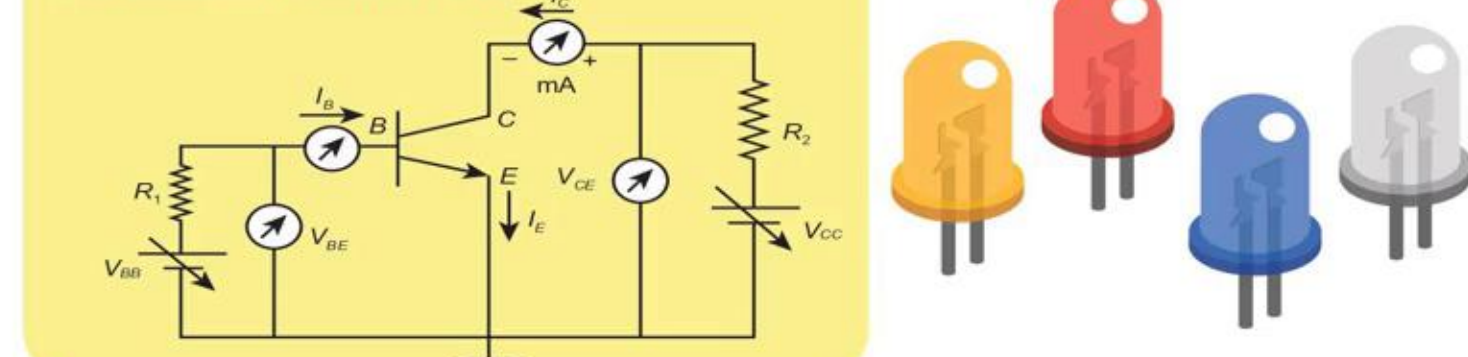
BASIC LOGIC GATES



WORKING OF N - P - N TRANSISTOR



CE CONFIGURATION CHARACTERISATION

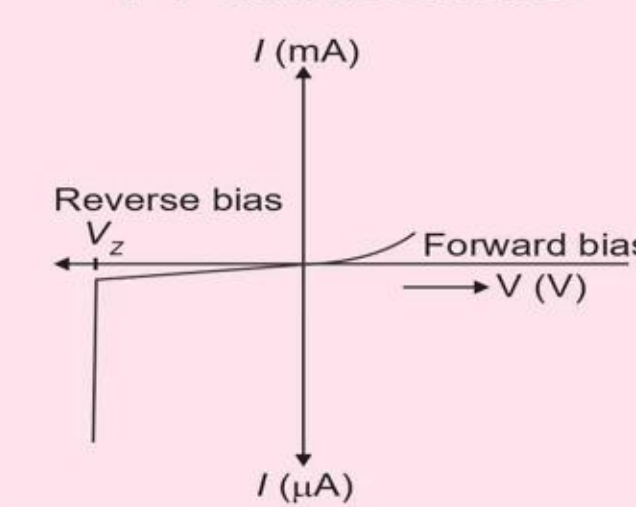


SEMICONDUCTOR ZENER DIODES

Anode(A) Cathode(K)

Zener Diode

I-V characteristics



ZENER BREAKDOWN

- This phenomenon takes place in
- (1) P - N junction having 'high doping'
 - (2) P - N junction having thin depletion layer
- Here, P - N junction does not damage permanently

AVALANCHE BREAKDOWN

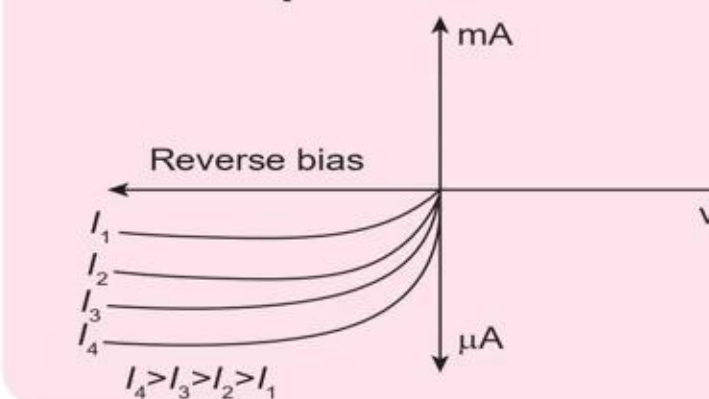
- This phenomenon takes place in
- (1) P - N junction having 'low doping'
 - (2) P - N junction having thick depletion layer.
- Here, P - N junction damages permanently due to abruptly increment of minorities during repetitive collisions.

PHOTODIODE

The symbol of photodiode is

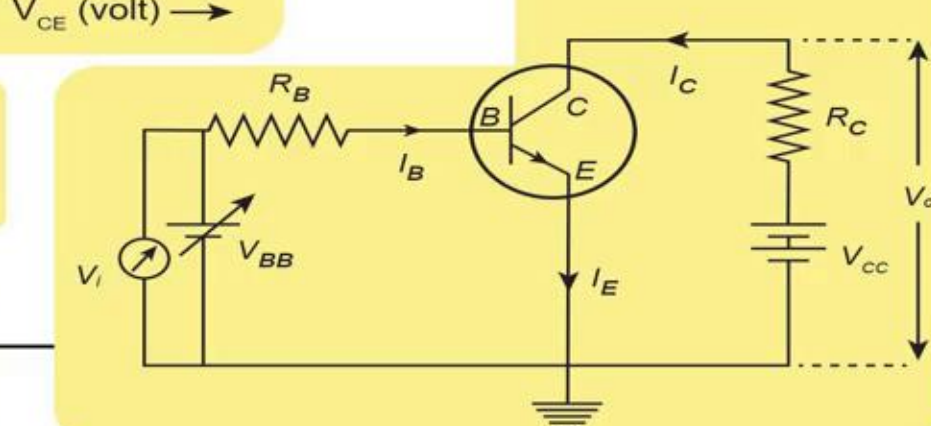


I-V characteristics of a photodiode

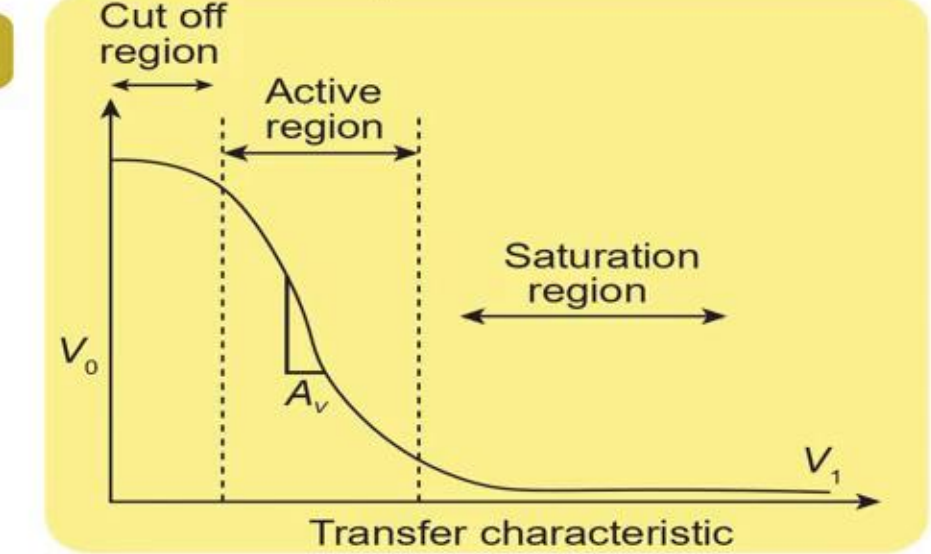
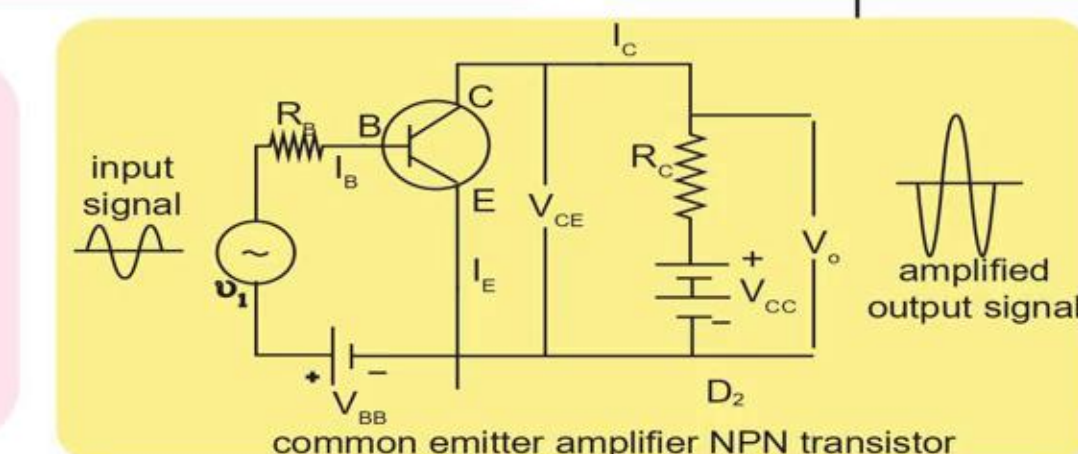


APPLICATION OF TRANSISTOR

AS A SWITCH

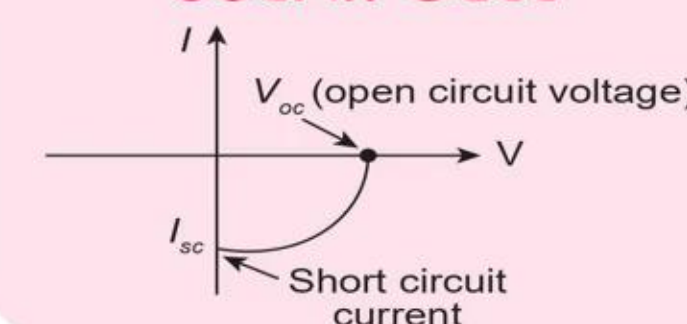


AS AN AMPLIFIER



I-V CHARACTERISTICS OF A SOLAR CELL.

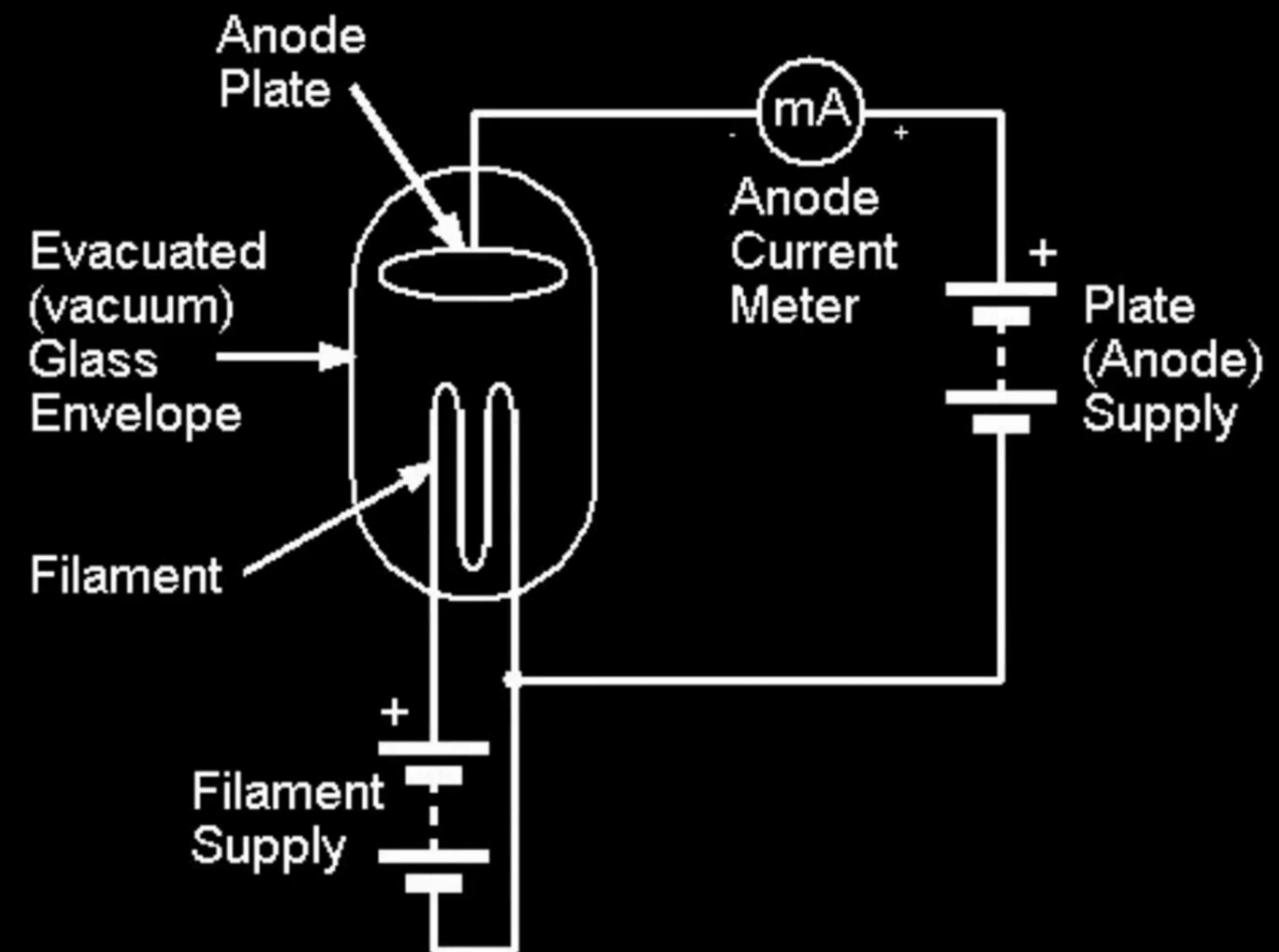
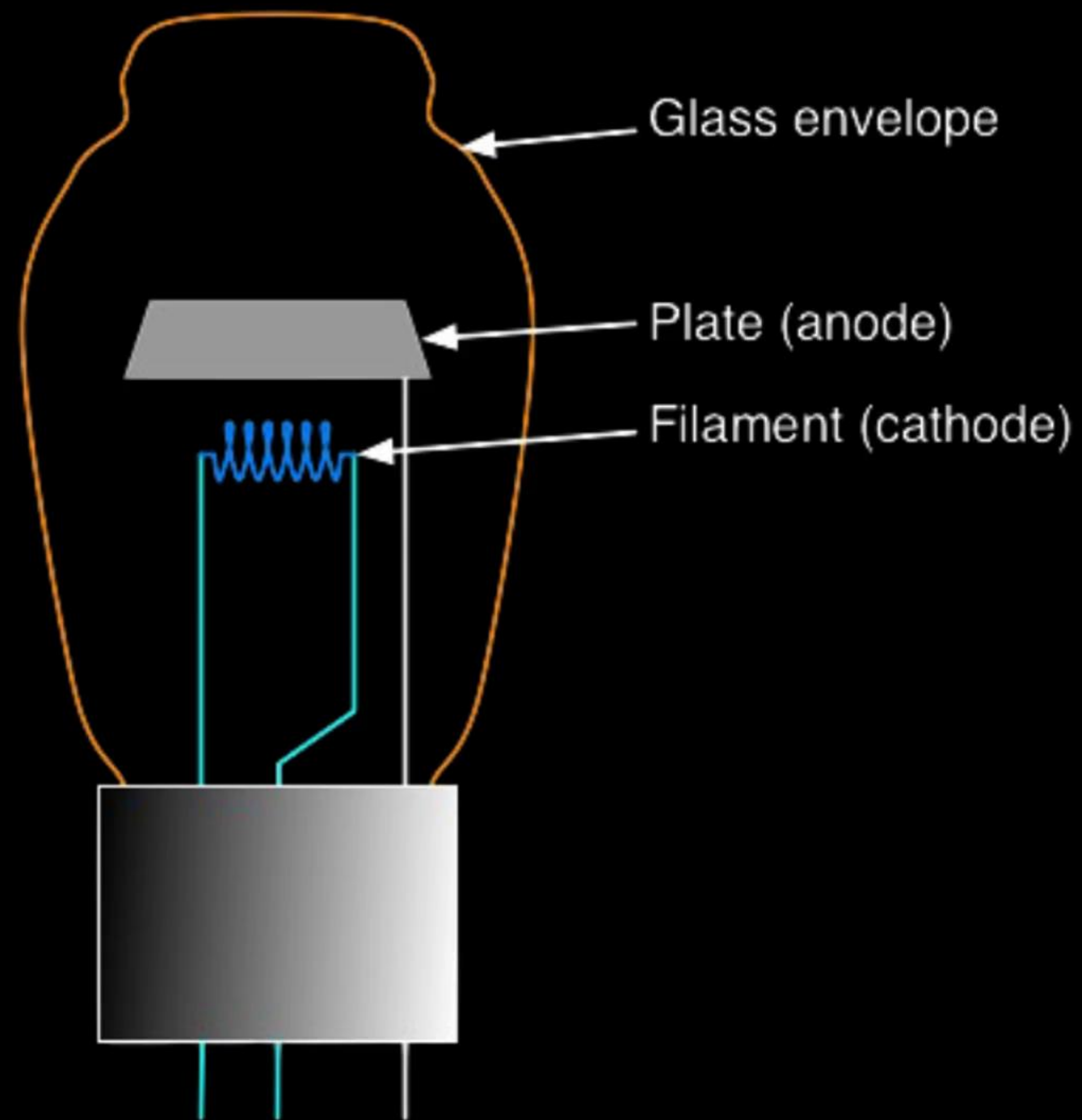
SOLAR CELL



Introduction

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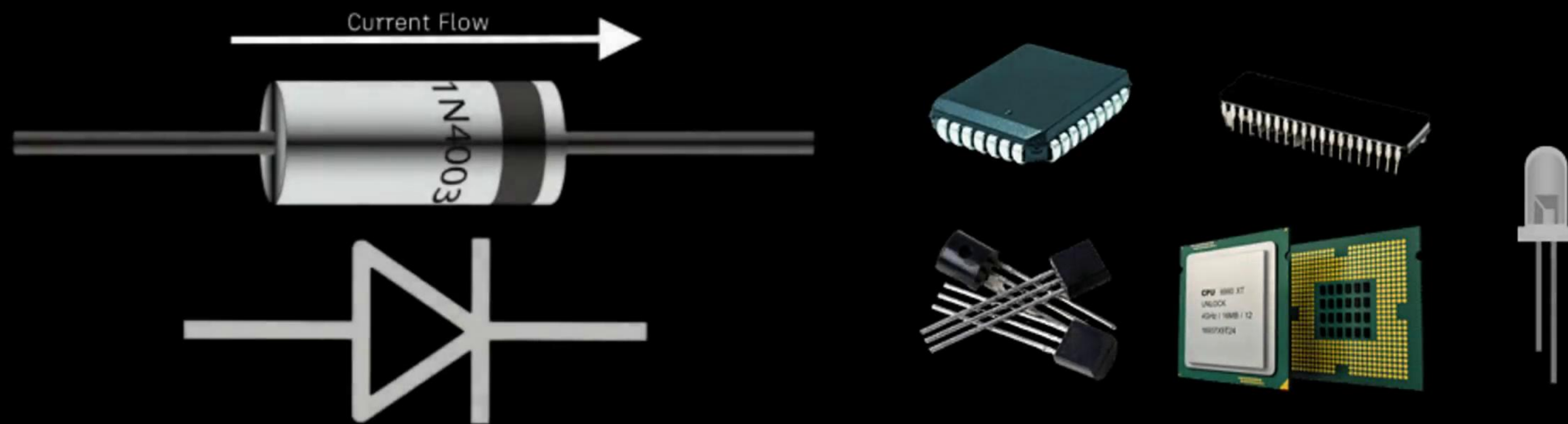


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- This chapter introduces **semiconductor physics, diodes, A few circuits illustrating their applications.**

Classification of Metals, Conductors and Semiconductors

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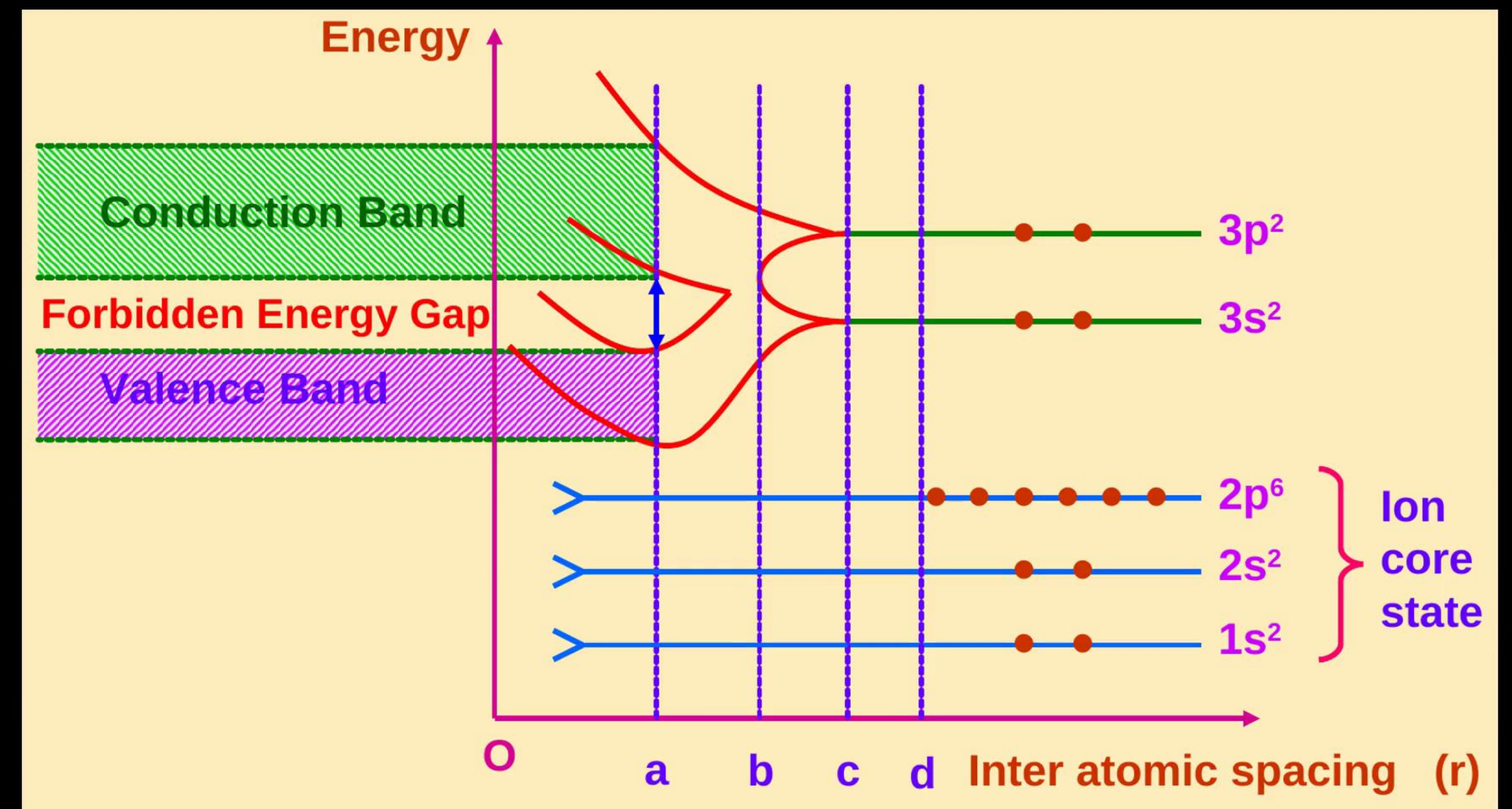
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- Order of conductivity: **Metals $\gg$ Semiconductors $\gg$ Insulators**.

Energy Band Formation

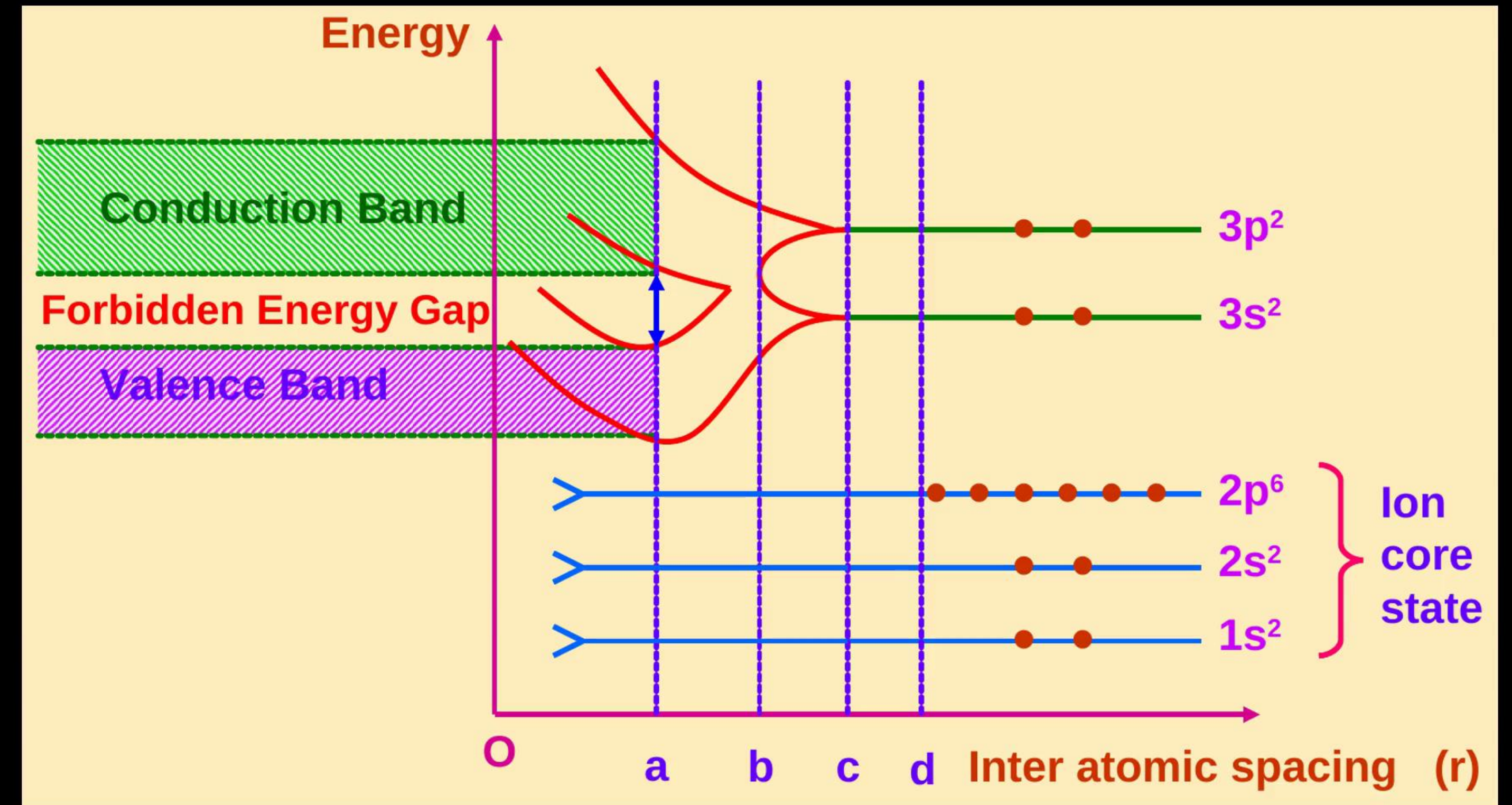
Energy Band Formation

- In an isolated atom, electrons occupy discrete energy levels.



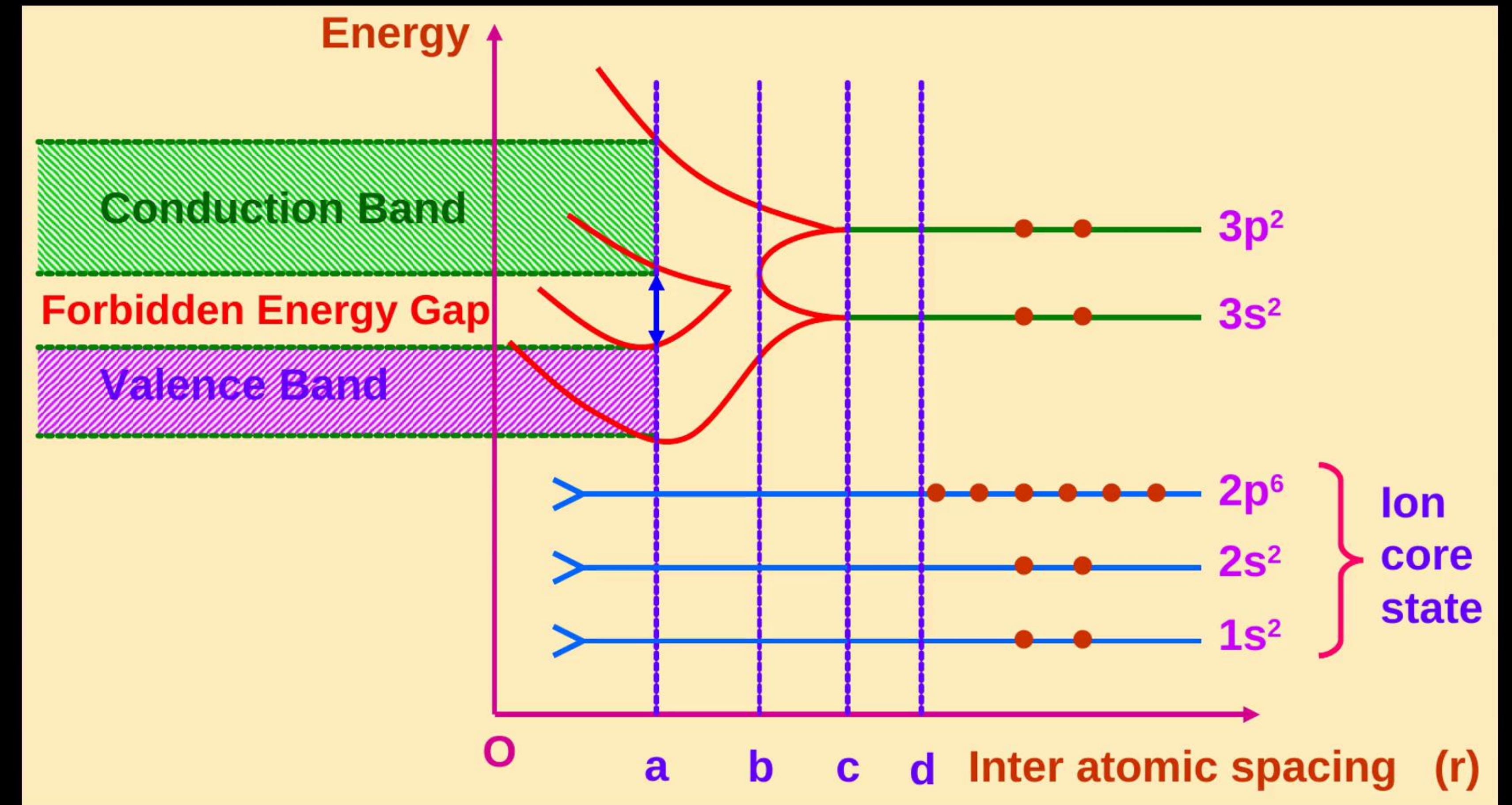
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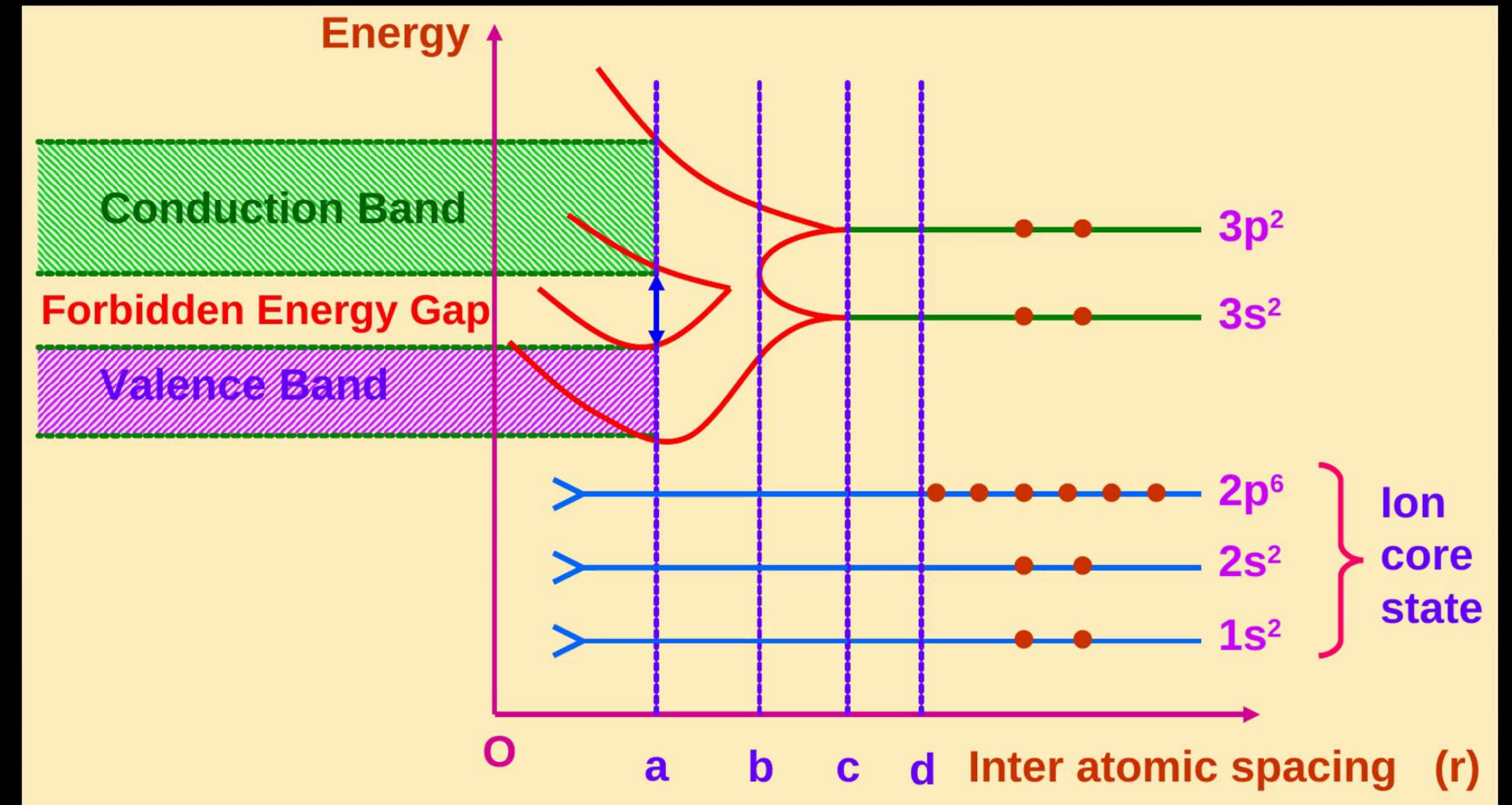
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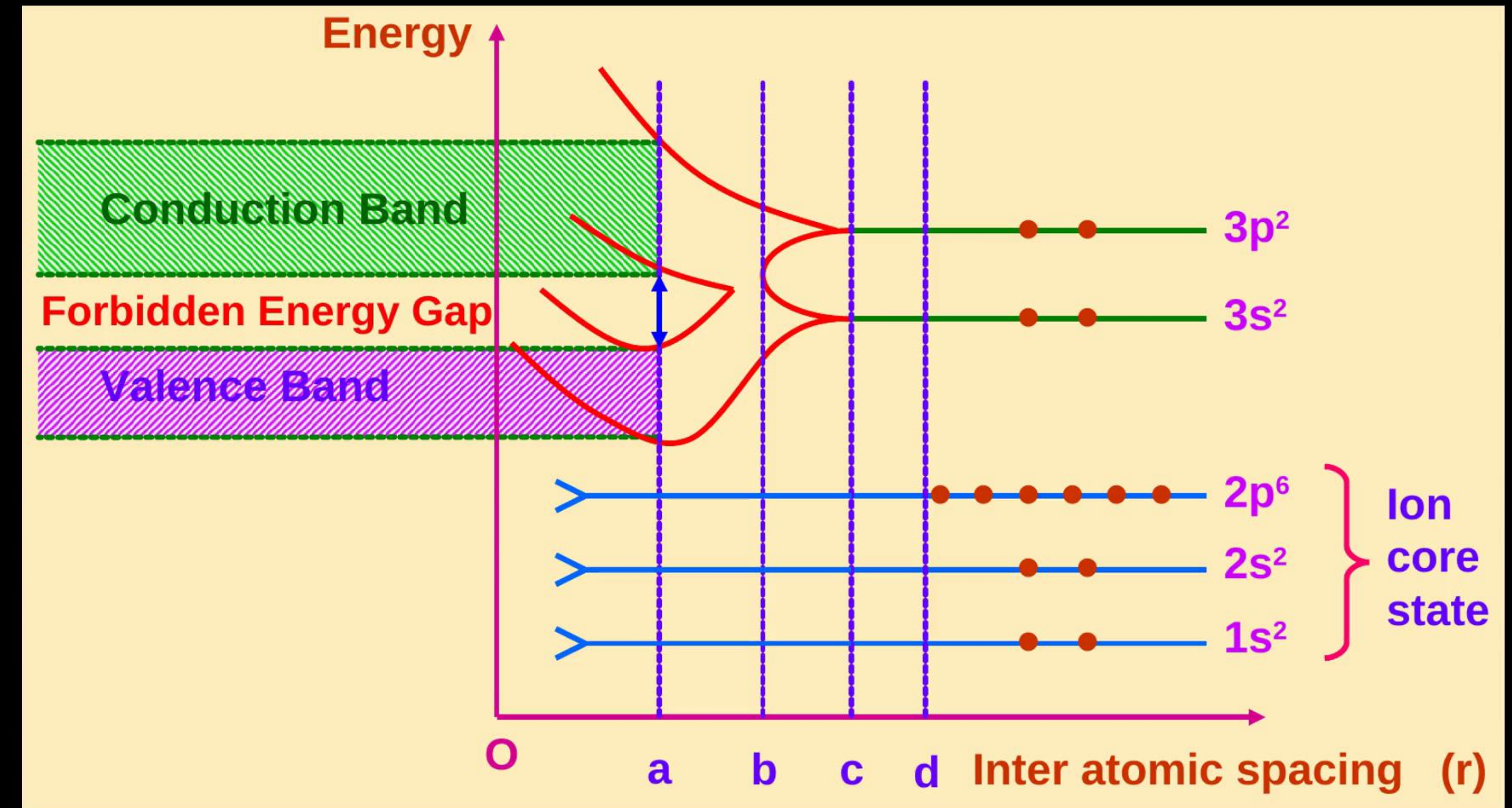
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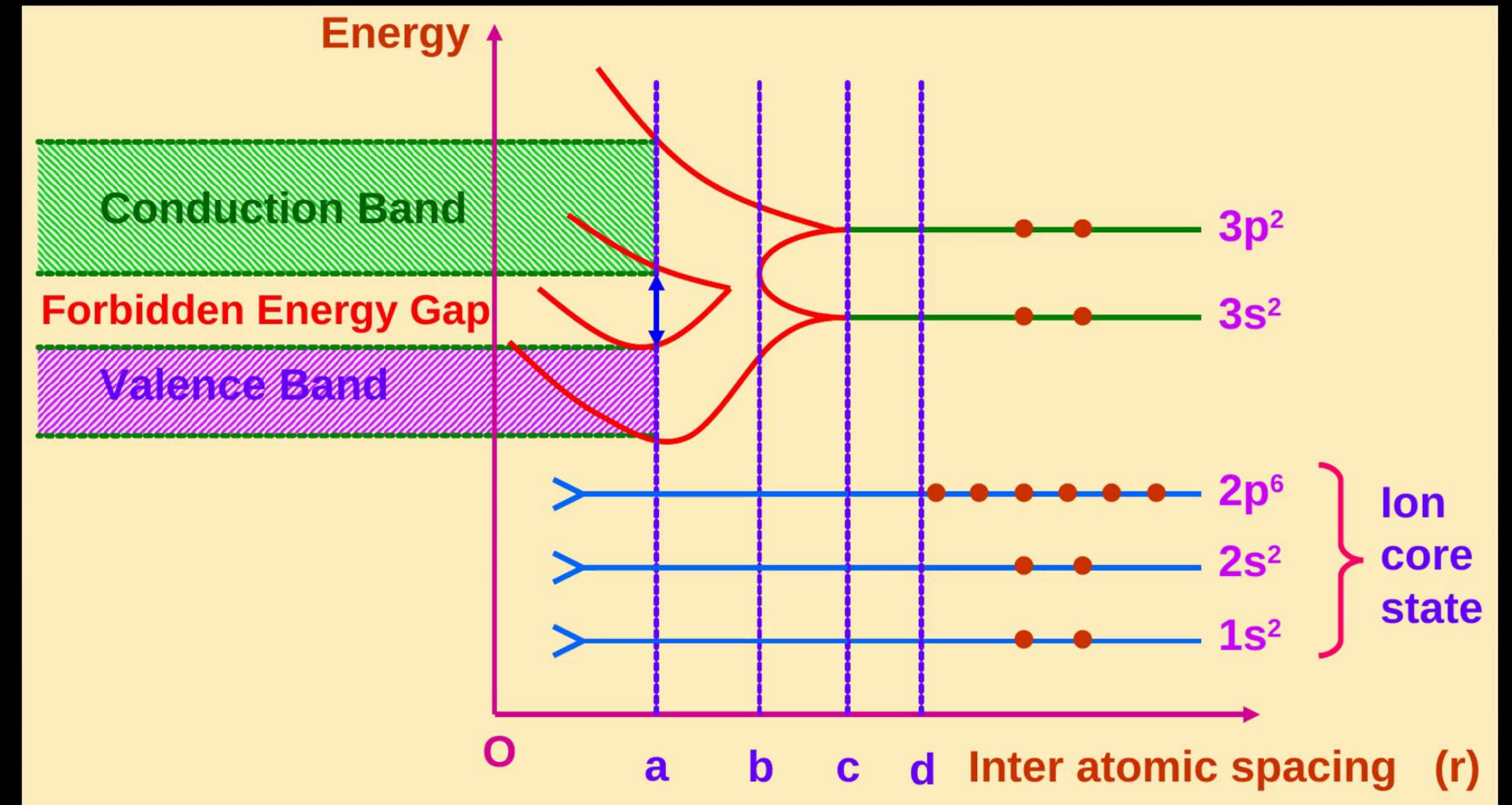
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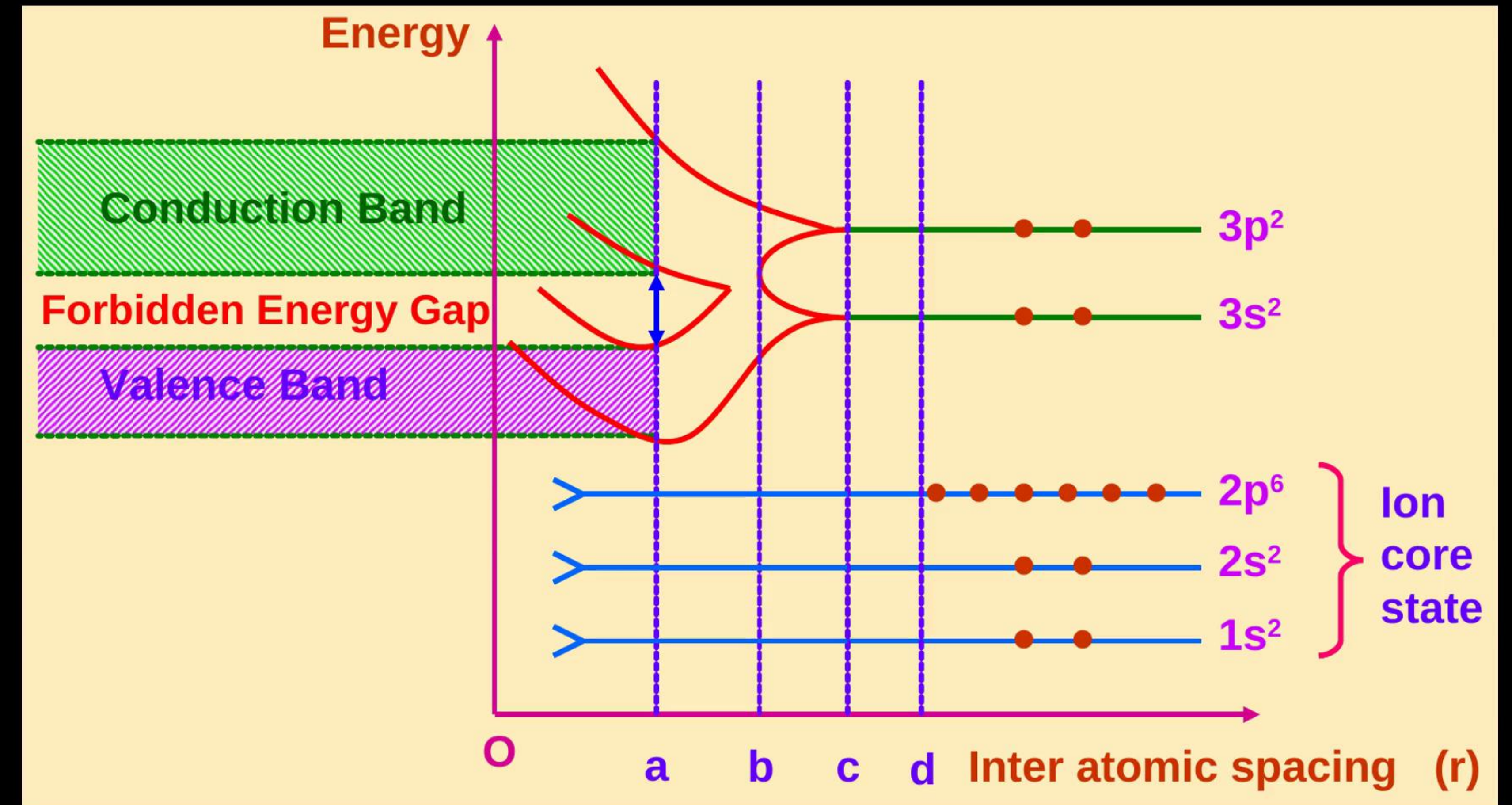
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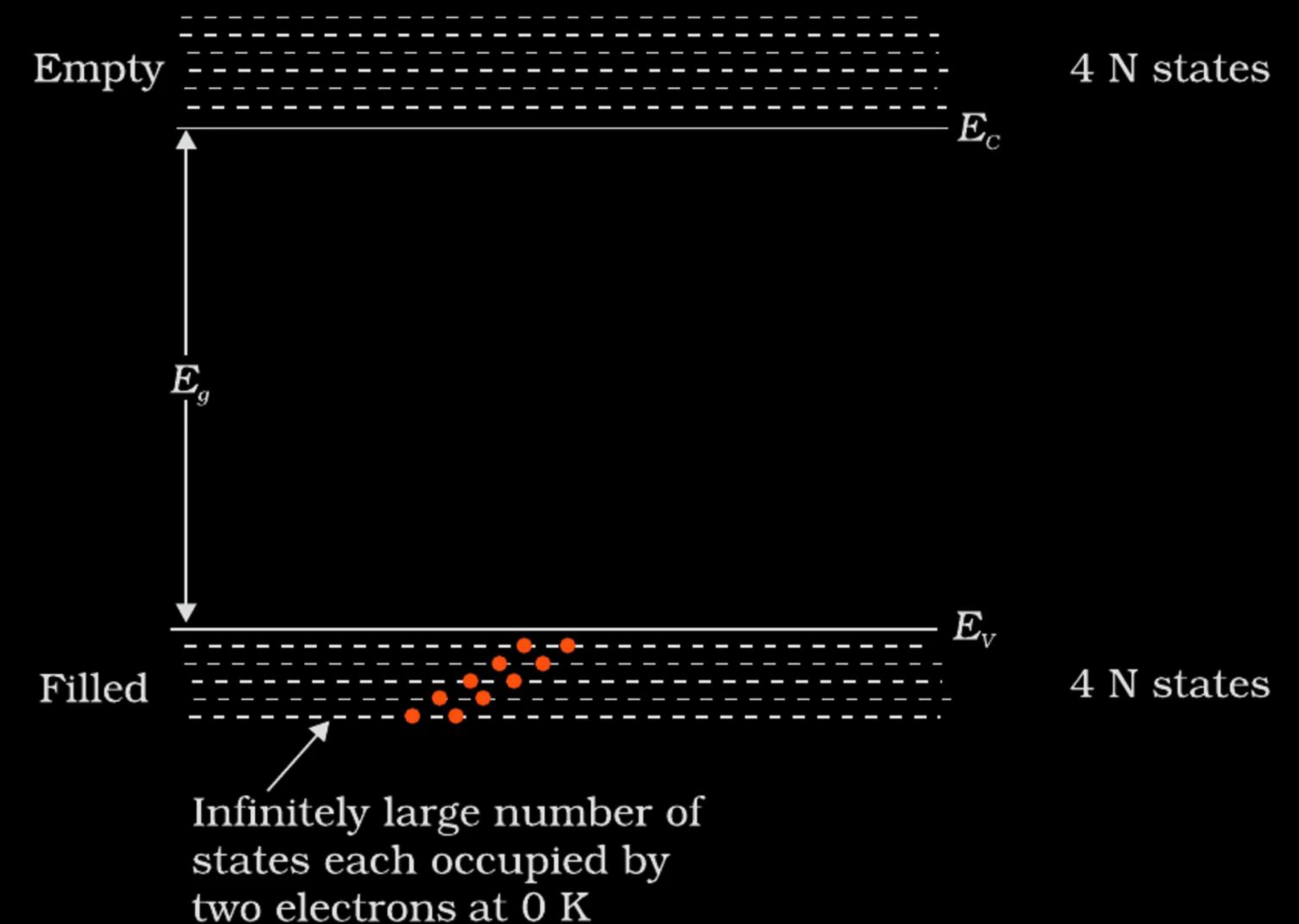


FIGURE 14.1 The energy band positions in a semiconductor at 0 K. The upper band, called the conduction band, consists of infinitely large number of closely spaced energy states. The lower band, called the valence band, consists of closely spaced completely filled energy states.

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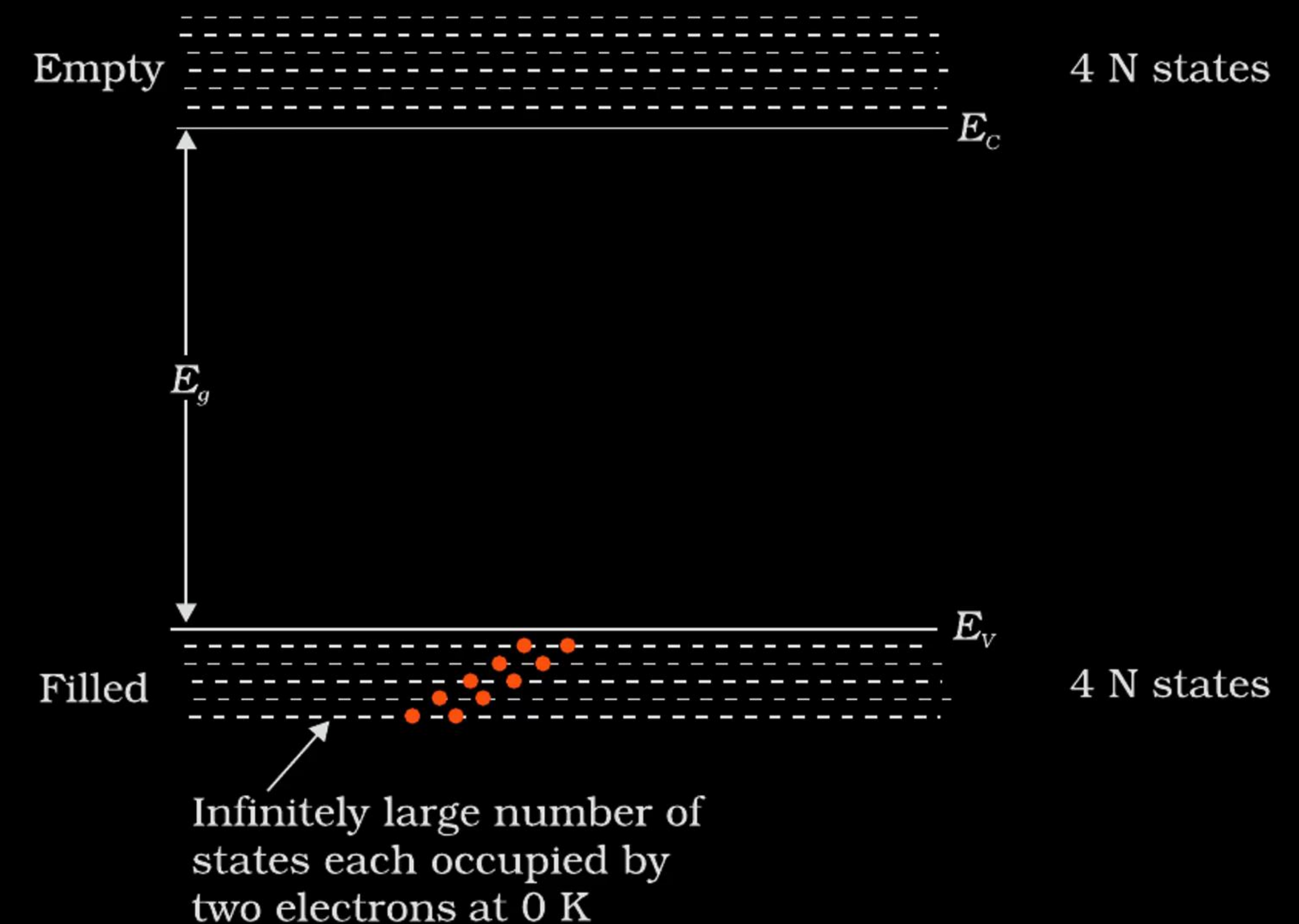


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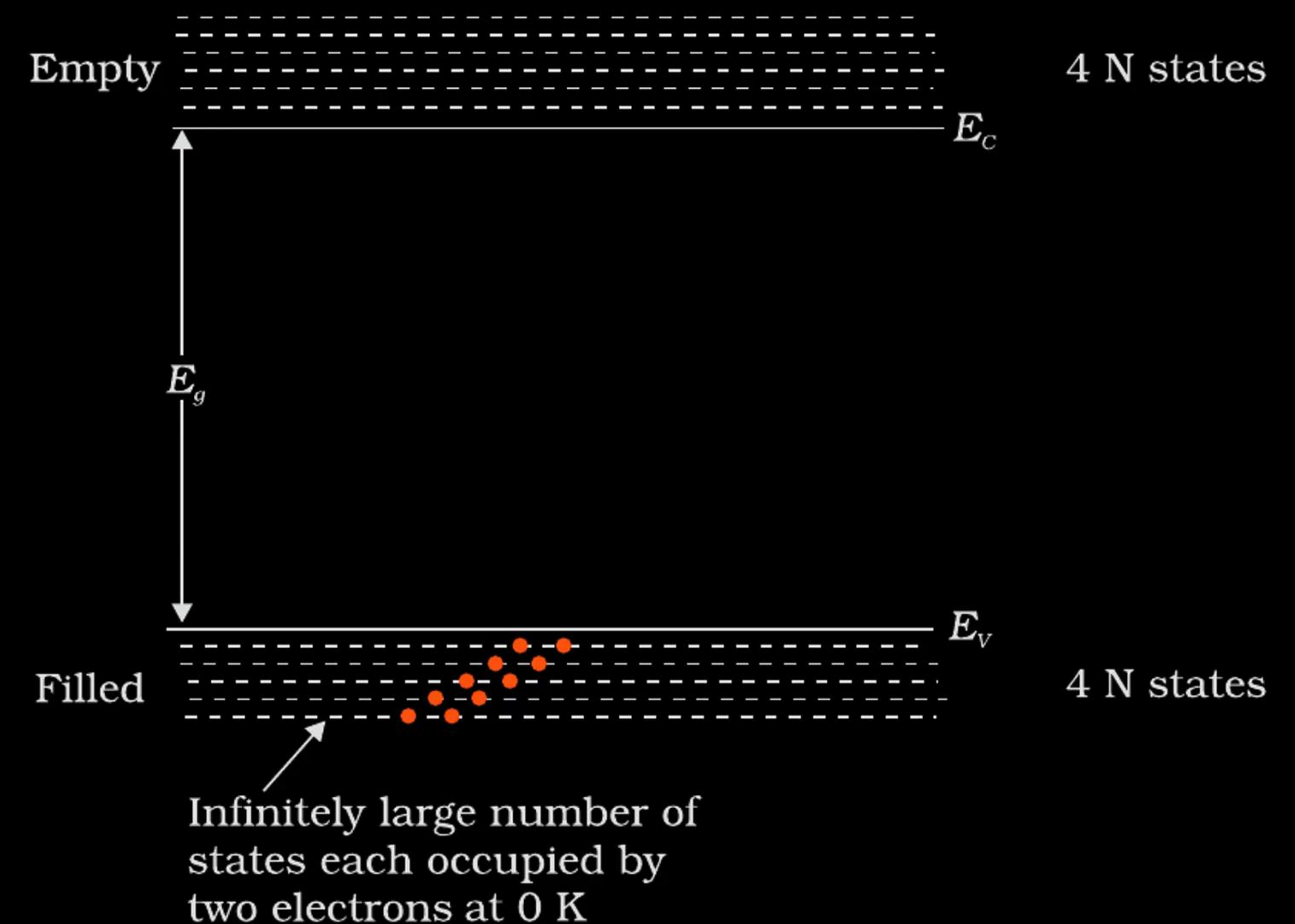


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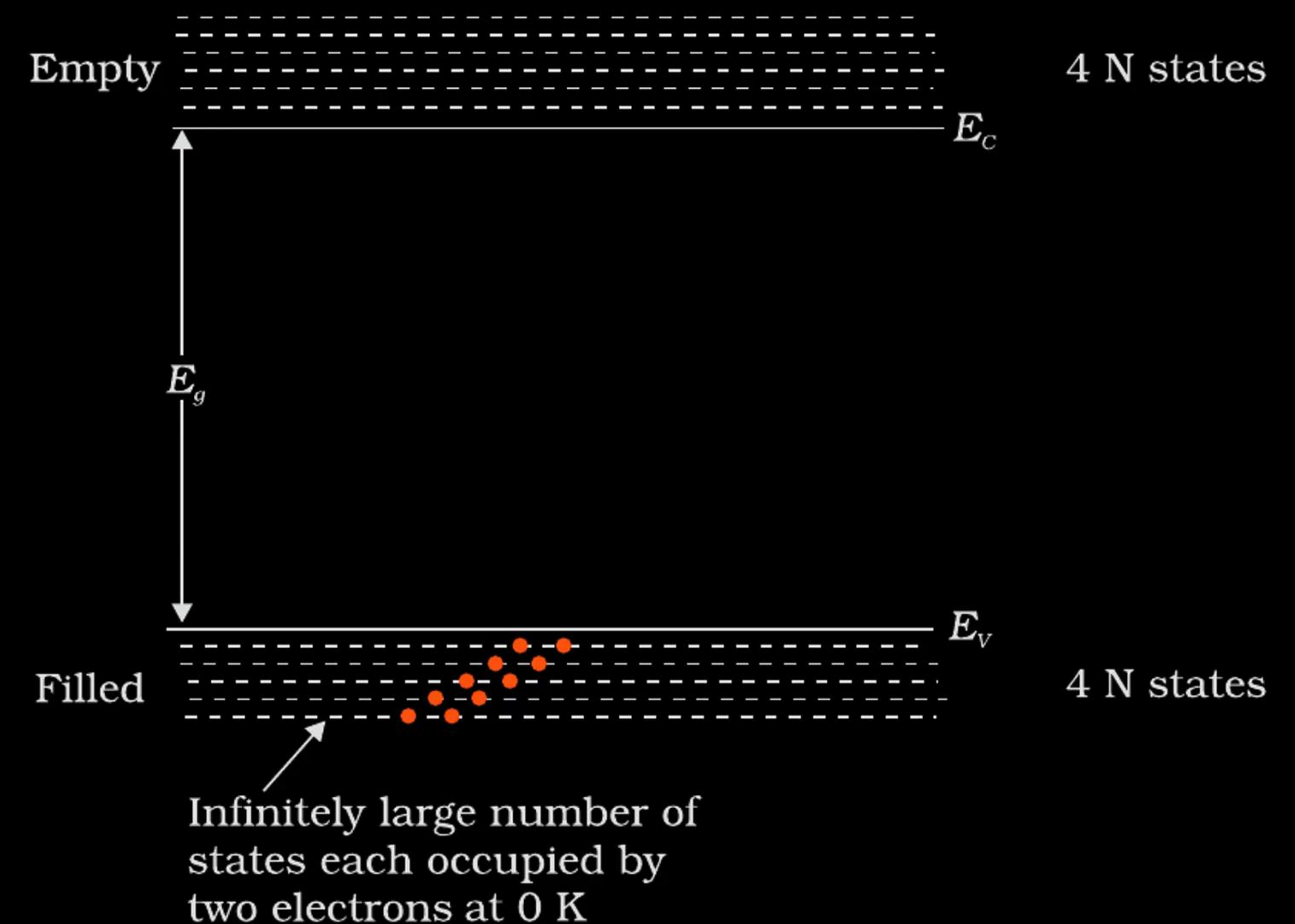
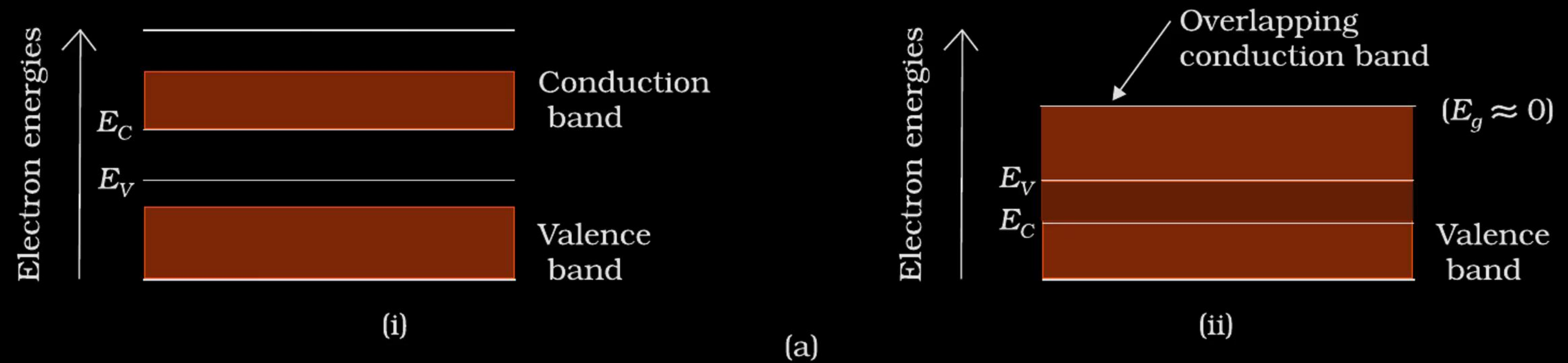


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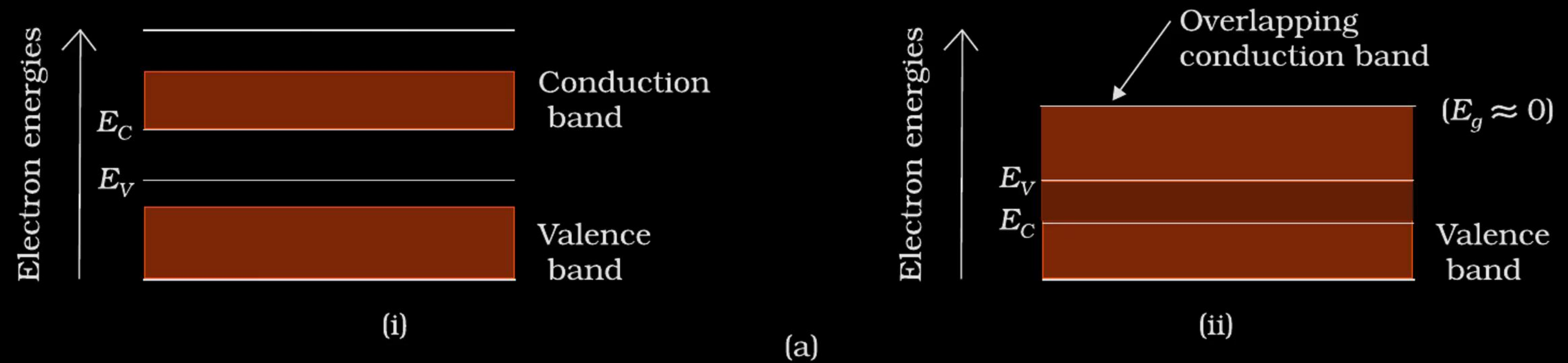
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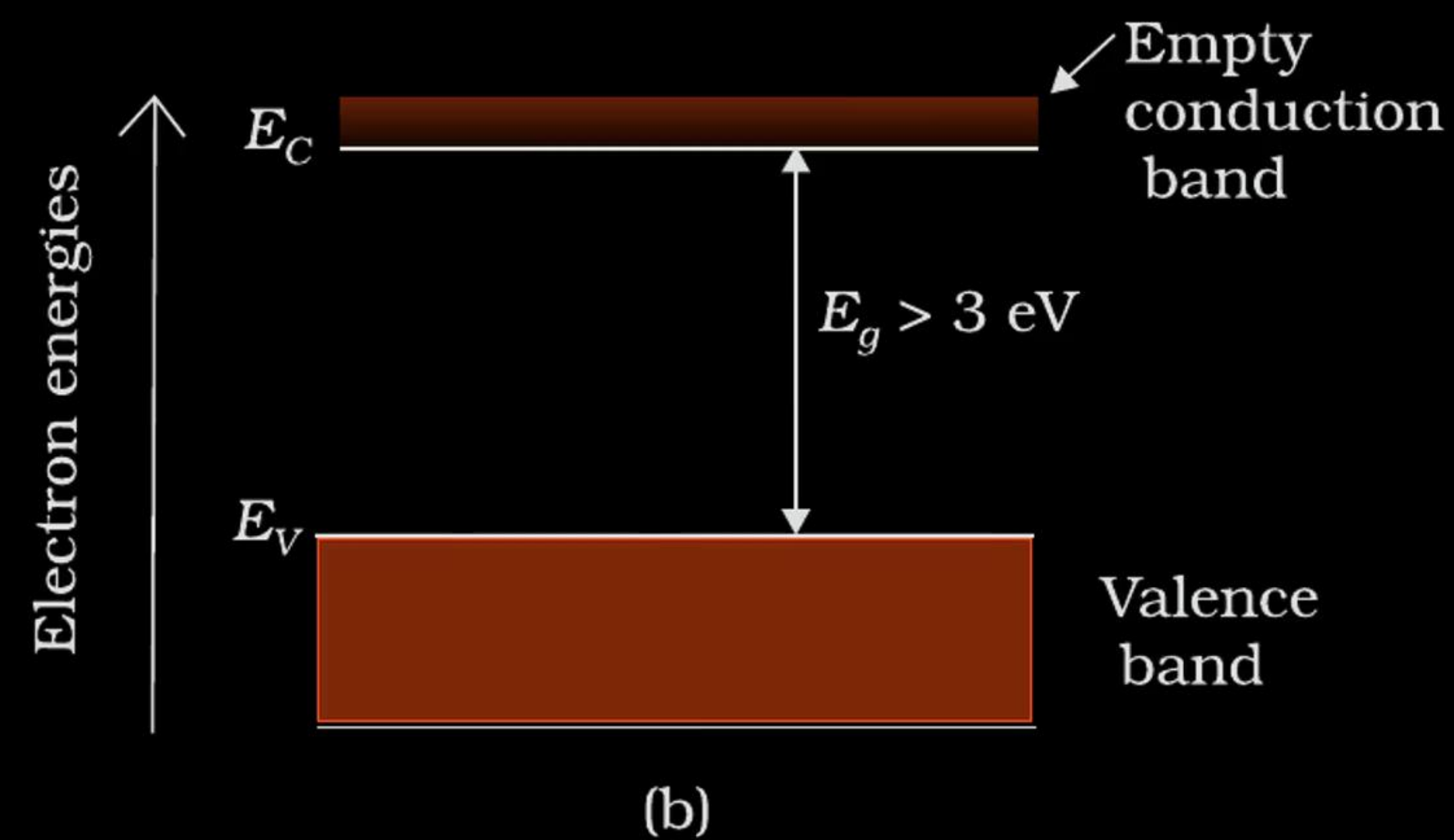


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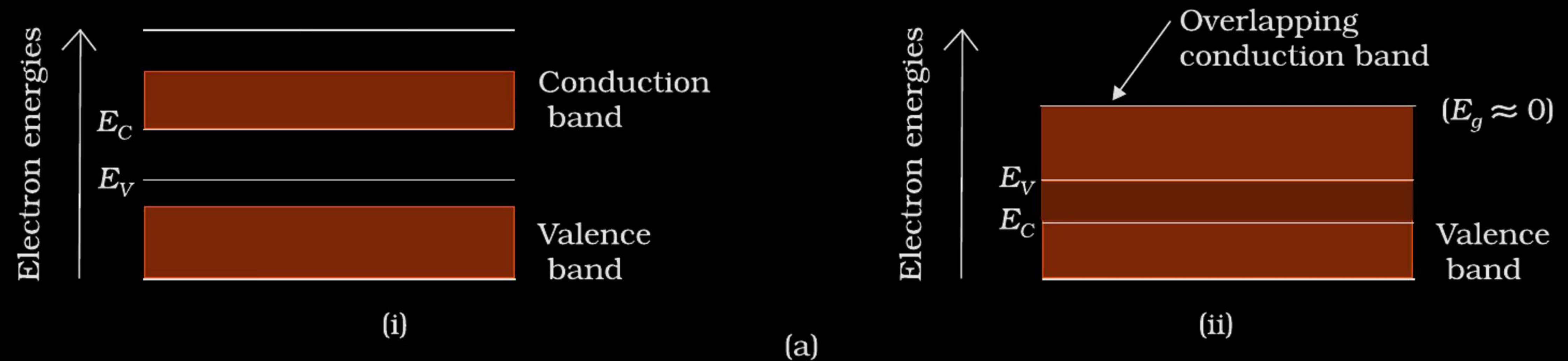


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- **Insulators:** Large energy gap $E_g > 3 \text{ eV}$ between VB and CB $\Rightarrow$ no electron flow.
- **Semiconductors:** Small gap $E_g < 3 \text{ eV}$; some electrons can jump from VB to CB at room temperature.

